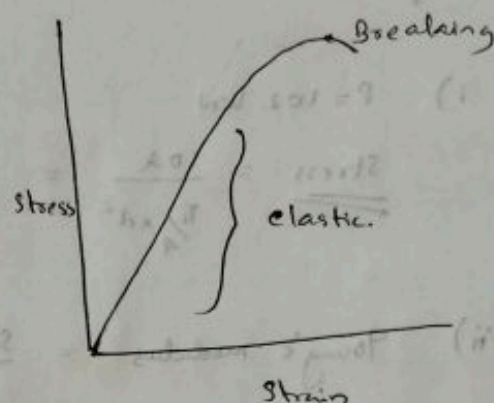
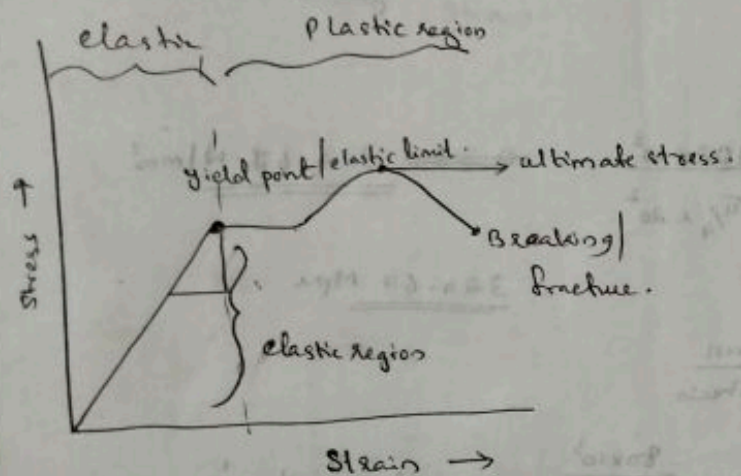


Module - 1

* Stress - strain Analysis

* Statically indeterminate problems

* Ductile , Brittle



* mild steel

Modulus of Elasticity / Yang's modulus = $\frac{\text{Stress}}{\text{Strain}}$ (slope)

$$\text{Stress} = \frac{\text{load}}{\text{Area}}$$

$$\text{Strain} = \frac{\Delta L}{L}$$

Stress at

$$\% \text{ elongation} = \frac{\text{Final length} - \text{Initial length}}{\text{Initial length}} \times 100$$

$$\% \text{ reduction in Area} = \frac{\text{Initial Area} - \text{Final area}}{\text{Initial Area}} \times 100$$

$$(2) \text{ Poisson's ratio} = \frac{\Delta D / D}{\Delta L / L}$$

$\Delta L = \text{final} - \text{initial}$

$$\frac{200}{0.25}$$

$$\Delta L = \frac{206.25 - 200}{200}$$

$$D_o = 20 \text{ mm}$$

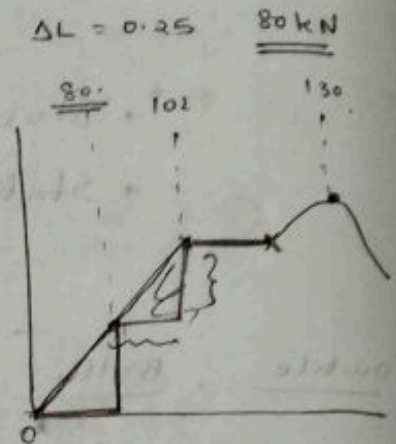
$$L_o = 200 \text{ mm}$$

Elastic limit load = 102 kN

Maximum load = 130 kN

$$\Delta L = 56 \text{ mm.}$$

$$D_f = 15 \text{ mm.}$$



i) $P = 102 \text{ kN}$

$$\text{Stress} = \frac{102}{\frac{\pi}{4} \times d^2} = \frac{102 \times 10^3}{\frac{\pi}{4} \times 20^2} = \underline{\underline{324.67 \text{ N/mm}^2}}$$

$$= \underline{\underline{324.67 \text{ MPa}}}$$

ii) Young's modulus = $\frac{\text{Stress}}{\text{Strain}}$

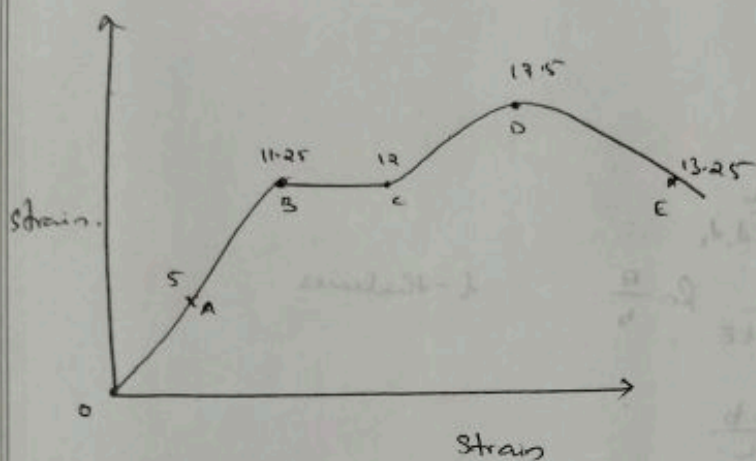
$$\text{Strain @ yield point} = \frac{\frac{80 \times 10^3}{\frac{\pi}{4} \times 20^2}}{\frac{0.25}{200}} = \underline{\underline{2 \times 10^5 \text{ N/mm}^2}}$$

iii) % elongation = $\frac{256 - 200}{200} = \underline{\underline{28\%}}$

iv) % reduction in area = $\frac{314.16 - 176.71}{314.16} = \underline{\underline{43.75\%}}$

$$\frac{\frac{\pi}{4} \times 20^2}{\frac{\pi}{4} \times 15^2}$$

v) ultimate tensile stress $\rightarrow \frac{130 \times 10^3}{\frac{\pi}{4} \times 20^2} = \underline{\underline{413.8 \text{ MPa}}}$



	Stress	Strain
A	5	5
B	11.25	11.25
C	11.25	12
D	17.5	17.5
E	13.25	23.25

$$\frac{11.25}{5} = \frac{\text{ultimate stress}}{\text{initial stress}} = (H) \text{ stress increase}$$

$$\frac{1}{H} = 0.4$$

$$\frac{17.5}{11.25} = \frac{\text{ultimate stress}}{\text{initial stress}} = (H) \text{ stress increase}$$

$$\frac{1}{H} = 0.6$$

$$\Delta L = \frac{PL}{AE}$$

Circular Tape $\Rightarrow \Delta L = \frac{4PL}{\pi E d_1 d_2}$

Rectangular Tape $\Rightarrow \Delta L = \frac{P}{k t E} \ln \frac{B}{b}$

t - thickness

$$k = \frac{B-b}{L}$$

* Same force $\Rightarrow$ same elongation.

Temperature $\Rightarrow \Delta L = L \alpha t$

* Heat $\Rightarrow$ Forces are same

$$\text{Poisson's ratio } (\mu) = \frac{\text{Lateral Strain}}{\text{Linear strain}} = \frac{\Delta d/d}{\Delta l/l}$$

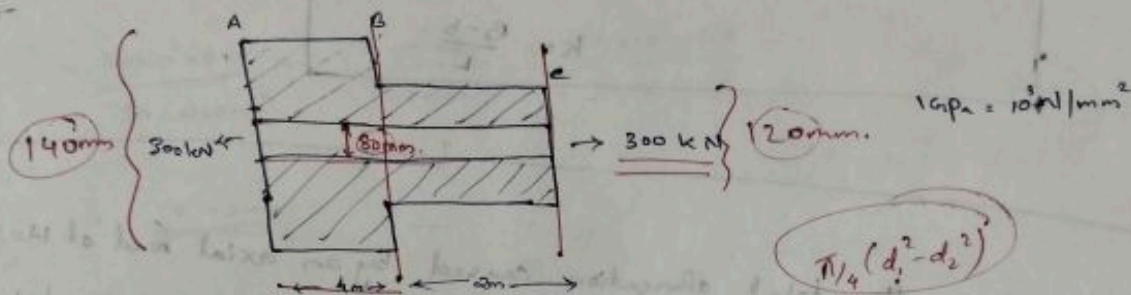
$$\text{Modulus of rigidity, } G = \frac{mE}{2(m+1)}$$

$$\left. \vphantom{\frac{mE}{2(m+1)}} \right\} m = \frac{1}{\mu}$$

$$\text{Bulk modulus, } k = \frac{mE}{3(m-2)}$$

Module - 1

- ① A 6m long hollow bar of circular section has 140 mm diameter for a length of 4m, while it has 120 mm diameter for a length 2m. The bore diameter is 80 mm. Throughout. Find elongation of bar. Take Young's modulus $E = 200 \text{ GPa}$.



Elongation, $\Delta L = \frac{PL}{AE}$

$\frac{PL_1}{A_1 E} + \frac{PL_2}{A_2 E}$

$P = 300 \times 10^3 \text{ N}$

$E = 200 \times 10^3 \text{ N/mm}^2$

$L = 4 \times 10^3 \text{ mm} + 2 \times 10^3 \text{ mm}$

$A = \frac{\pi}{4} d^2$

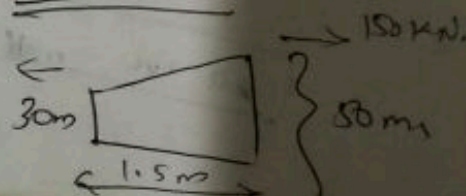
Hollow bar $\Rightarrow A = \frac{\pi}{4} (d_1^2 - d_2^2)$

$$\Delta L = \frac{300 \times 10^3 \times 4 \times 10^3}{\frac{\pi}{4} (140^2 - 80^2) \times 200 \times 10^3} + \frac{300 \times 10^3 \times 2 \times 10^3}{\frac{\pi}{4} (120^2 - 80^2) \times 200 \times 10^3}$$

$= 1.056 \text{ mm}$

- ② A Rod tapers uniformly from 50mm diameter to 30mm diameter in a length of 1.5m is subjected to an axial force of 150 kN.

Determine the elongation of the rod. $E = 200 \text{ GPa}$



In case of Taper.

• Circular taper $\rightarrow \Delta L = \frac{4PL}{\pi E d_2 d_1} \checkmark$
 • Rectangular taper $\rightarrow \Delta L = \frac{P}{k t E} \ln \frac{B}{b}$
 $k = \frac{B-b}{L}$

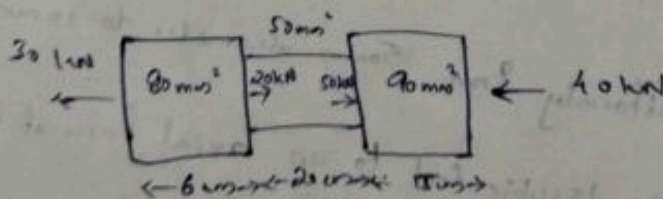
t = thickness of taper
 B, b = widths

- ③ Compute the total elongations caused by an axial load of 120 kN applied to a flat bar of 22 mm thickness tapering from width of 100 mm to 30 mm in a length of 8 m. $E = 210 \text{ Gpa}$.

$$k = \frac{B-b}{L} = \frac{100-30}{8000} = 0.00875$$

$$\Delta L = \frac{120 \times 10^3}{0.00875 \times 22 \times 210 \times 10^3} \ln \frac{100}{30} = 3.5739 \text{ mm}$$

- ④ A stepped bar is loaded as shown. Determine total extension of the bar. $E = 200 \text{ Gpa}$.



left +ve, right negative

$$P_1 = 30 \text{ kN}$$

$$P_2 = 30 - 20 = 10 \text{ kN}$$

$$P_3 = 30 - 20 - 50 = -40 \text{ kN}$$

$$P_4 = 30 - 20 - 50 + 40 = 0$$

$$\Delta L = \frac{P_1 L_1}{A_1 E} + \frac{P_2 L_2}{A_2 E} + \frac{P_3 L_3}{A_3 E}$$

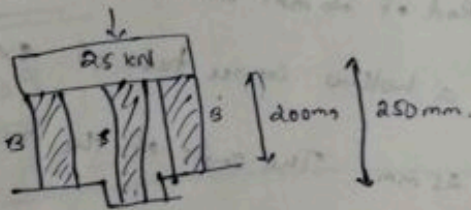
$$= \frac{30 \times 10^3 \times 60}{80 \times 200 \times 10^3} + \frac{10 \times 10^3 \times 200}{50 \times 200 \times 10^3} + \frac{-40 \times 10^3 \times 150}{40 \times 200 \times 10^3}$$

$$= \underline{\underline{-0.02 \text{ mm}}}$$

$$\Delta L = \frac{PL}{AE} = \frac{PL}{\frac{\pi d_1 d_2 E}{4}}$$

$$= \frac{4PL}{\pi d_1 d_2 E}$$

- ⑤ A steel rod of cross sectional area 800 mm^2 and ~~two~~ two brass rods each of cross sectional area 500 mm^2 together support a load of 25 kN . Calculate the stresses in the rod. $E_s = 200 \text{ GPa}$ $E_b = 100 \text{ GPa}$.



same force $\Rightarrow$ same elongation.

$$\Delta L_s = \Delta L_b$$

$$\frac{P_s L_s}{A_s E_s} = \frac{P_b L_b}{A_b E_b}$$

$$\sigma_s \frac{L_s}{E_s} = \sigma_b \frac{L_b}{A_b}$$

$$\frac{P}{A} = \sigma$$

$$\sigma_s \times \frac{250}{200 \times 10^3} = \sigma_b \times \frac{200}{100 \times 10^3}$$

$$\Rightarrow 1.25 \times 10^3 \sigma_s = 2 \times 10^3 \sigma_b$$

$$\Rightarrow 1.25 \sigma_s = 2 \sigma_b$$

$$2\sigma_B - 1.25\sigma_S = 0 \quad \text{--- (1)}$$

$$2P_B + P_S = P$$

↓

$$2\sigma_B A_B + \sigma_S A_S = 25000$$

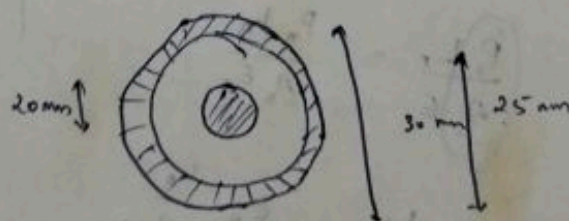
$$2 \times 500 \times \sigma_B + \sigma_S \times 800 = 25000$$

$$1000\sigma_B + 800\sigma_S = 25000 \quad \text{--- (2)}$$

$$\Rightarrow \sigma_B = 10.96 \text{ N/mm}^2$$

$$\sigma_S = 17.84 \text{ N/mm}^2$$

- ⑥ A mild steel Rod of 20 mm diameter and 300 mm long is enclosed centrally inside a hollow copper tube of external diameter 30 mm and internal diameter 25 mm. The ends of the rod and tube are brazed together and the composite bar is subjected to an axial pull of 40 kN as shown. Find stress developed in rod and tube. Also find change in length of steel rod. $E_S = 200 \text{ GPa}$ $E_C = 100 \text{ GPa}$.



$$\Delta L_s = \Delta L_c$$

$$\sigma_s \frac{L_s}{E_s} = \sigma_c \frac{L_c}{E_c}$$

$$L_s = L_c \Rightarrow \frac{\sigma_s}{200 \times 10^3} = \frac{\sigma_c}{100 \times 10^3}$$

$$\sigma_s = 2\sigma_c \Rightarrow 2\sigma_c - \sigma_s = 0 \quad \text{--- (1)}$$

$$P_s + P_c = 40 \times 10^3$$

$$\sigma_s \times A_s + \sigma_c \times A_c = \sigma_s \times \frac{\pi}{4} 20^2 + \sigma_c \times \frac{\pi}{4} (30^2 - 25^2)$$

$$= 314 \sigma_s + 215.98 \sigma_c = 40000$$

$$= 314 \sigma_s + 215.98 \sigma_c - 40000 = 0 \quad \text{--- (2)}$$

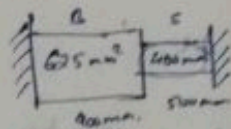
$$\sigma_s = \underline{\underline{94.78 \text{ N/mm}^2}}$$

$$\sigma_c = \underline{\underline{47.39 \text{ N/mm}^2}}$$

- 7) A bar made of Brass and Steel as shown is held between 2 rigid supports A and C. Find the stresses in each material if the temperature rises by 40° . $E_b = 1 \times 10^5 \text{ N/mm}^2$ $E_s = 2 \times 10^5 \text{ N/mm}^2$
- $\alpha_b = 19 \times 10^{-6}/^\circ\text{C}$ $\alpha_s = 12 \times 10^{-6}/^\circ\text{C}$

Temperature $\rightarrow$

$$\Delta L = L \alpha t$$



$$\Delta L = \Delta L_B + \Delta L_C$$

$$= 400 \times 12 \times 10^{-6} \times 40 + 500 \times 12 \times 10^{-6} \times 40$$

$$= 0.304 + 0.24 = \underline{\underline{0.54}}$$

$$\Delta L = \frac{PL}{AE} + \frac{PL}{AE}$$

$$0.54 = \sigma_S \frac{L_S}{E_S} + \sigma_B \frac{L_B}{E_B}$$

$$0.54 = \sigma_S \times 250 \times 10^{-5} + \sigma_B \times 400 \times 10^{-5} \quad \text{--- (1)}$$

heat $\Rightarrow$ Force same

$$P_S = P_B$$

$$\sigma_S A_S = \sigma_B A_B$$

$$400 \sigma_S = 625 \sigma_B \quad \text{--- (2)}$$

$$\sigma_S = 107.42 \text{ N/mm}^2$$

$$400 \sigma_S = 625 \sigma_B$$

$$\sigma_S = \frac{625 \sigma_B}{400} \quad \text{--- (3)}$$

$$\sigma_B = 68.26 \text{ N/mm}^2$$

A steel tube of 35 mm outer diameter and 30 mm inner diameter encloses a gunmetal rod of 25 mm dia and is rigidly joined at each end. It is at a temperature of 40°C . There is no longitudinal stress. Determine the stress developed in the rod and the tube when the temperature of the assembly is raised to 240°C .

$$E_s = 205 \text{ GPa} \quad E_g = 91.5 \text{ GPa} \quad \alpha_s = 11 \times 10^{-6}/^{\circ}\text{C} \quad \alpha_g = 18 \times 10^{-6}/^{\circ}\text{C}$$

$$\text{Tube shaft + temp} \Rightarrow P = \frac{t(\alpha_g - \alpha_s)}{\frac{1}{A_s E_s} + \frac{1}{A_g E_g}}$$

$$\sigma = P/A$$

$t = \text{temp difference}$

$$= 240 - 40$$

$$P = \frac{200 (18 \times 10^{-6} - 11 \times 10^{-6})}{\frac{1}{\frac{\pi}{4} (35^2 - 30^2) \times 205 \times 10^9} + \frac{1}{\frac{\pi}{4} 25^2 \times 91.5 \times 10^9}}$$

$$= \underline{\underline{33841 \text{ N}}}$$

$$\sigma = P/A$$

$$\sigma_s = \frac{P}{\frac{\pi}{4} (35^2 - 30^2)}$$

$$= \underline{\underline{132.57 \text{ N/mm}^2}}$$

$$\sigma_g = \frac{P}{\frac{\pi}{4} 25^2}$$

$$= \underline{\underline{68.9 \text{ N/mm}^2}}$$

- ④ Two parallel walls, 8m apart are to be stayed together by a steel rod of 30mm dia with the help of washers and nut at the ends. The steel rod is passed through the metal plates and is heated when its temperature is raised to 90°C , the nuts are tightened. Determine the pull in the bar when it is cooled to 24°C if

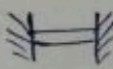
ends is 2 mm.

$$E = 205 \text{ GPa}$$

$$\alpha_s = 11 \times 10^{-6} / ^{\circ}\text{C}$$

② $\Rightarrow$

$$\Delta L = L \alpha t$$

ends do not yield 

$$\Delta L = \frac{PL}{AE} = \frac{\sigma L}{E}$$

$\Rightarrow$ both end fixed

$$\Rightarrow L \alpha t = \frac{\sigma L}{E}$$

$$\Rightarrow \alpha t = \frac{\sigma}{E}$$

$$\sigma = \alpha t E$$

$$t = 90 - 24$$

$$= 66$$

$$= 11 \times 10^{-6} \times 205 \times 10^3 \times 66$$

$$= 148.83 \text{ N/mm}^2$$

$$\text{Pull, } P = \sigma A = 148.83 \times \frac{\pi}{4} \times 30^2$$

$$= 105201.73 \text{ N}$$

total yielding at ends; 2mm. (gap)

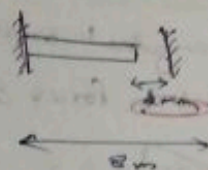
$$\frac{PL}{AE} = Lxt - a.$$

$$\sigma \frac{L}{E} = Lxt - a$$

$$\sigma = \frac{(Lxt - a)E}{L} = \frac{(8000 \times 11 \times 10^{-6} \times 66 - 2) \times 205 \times 10^3}{8000}$$

$$\sigma = 97.56 \text{ N/mm}^2$$

$$\text{Pull, } P = \sigma \cdot A = 68975.24 \text{ N}$$



known

$L = 8000$

$$\Delta L = Lxt - a$$

In an experiment, a bar of 30 mm dia is subjected to pull of 60 kN. The measured extension on a gauge length of 200 mm is 0.09 mm and change in diameter is 0.0039 mm. Calculate poisson ratio and Bulk modulus 'K' and modulus of rigidity 'G'.

$$\text{Poisson ratio } (\mu) = \frac{\text{lateral strain}}{\text{linear strain}} = \frac{\Delta d/d}{\Delta L/L} = \frac{0.0039/30}{0.09/200} = 0.288$$

$$G = \frac{\mu E}{2(\mu + 1)}$$

$$\Rightarrow \mu = \frac{1}{1.47} = 0.68$$

$$\Rightarrow E = \frac{60 \times 10^3 \times 200}{0.09 \times \pi/4 \times 30^2} = 188628.08 \text{ N/mm}^2$$

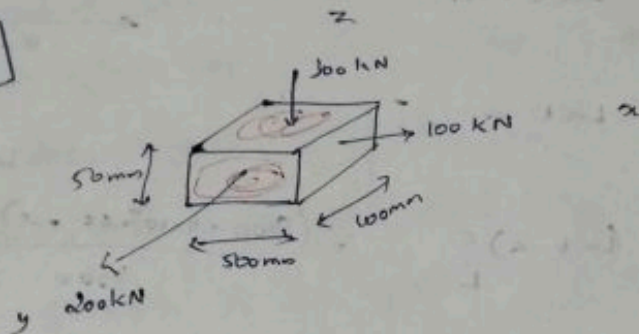
$$\Rightarrow G = 73167.39 \text{ N/mm}^2$$

$$= 149007.57 \text{ N/mm}^2$$

$$K = \frac{\mu E}{3(\mu - 2)}$$

- Q. A rectangular bar 500 mm long and 100 mm x 50 mm in cross section is subjected to forces as shown. Find change in volume of bar.

Change in volume



$$E = 200 \text{ GPa}$$

$$\mu = 0.25$$

Compression (-ve)

$$\sigma_x = \frac{P_x}{A_x} = \frac{100 \times 10^3}{100 \times 50} = 20 \text{ N/mm}^2$$

$$\epsilon_x = \frac{\sigma_x}{E} - \frac{\sigma_y}{mE} - \frac{\sigma_z}{mE} \quad \sigma_y = \frac{200 \times 10^3}{50 \times 50} = 8 \text{ N/mm}^2$$

$$\epsilon_y = \frac{\sigma_y}{E} - \frac{\sigma_x}{mE} - \frac{\sigma_z}{mE} \quad \sigma_z = -\frac{300 \times 10^3}{500 \times 100} = -6 \text{ N/mm}^2$$

$$\epsilon_z = \frac{\sigma_z}{E} - \frac{\sigma_x}{mE} - \frac{\sigma_y}{mE}$$

$$\epsilon_x = \frac{\sigma_x}{E} - \frac{\sigma_y}{mE} - \frac{\sigma_z}{mE}$$

$$\epsilon_y = \frac{\sigma_y}{E} - \frac{\sigma_x}{mE} - \frac{\sigma_z}{mE}$$

$$\epsilon_z = \frac{\sigma_z}{E} - \frac{\sigma_x}{mE} - \frac{\sigma_y}{mE}$$

$$= \frac{19.5}{E}$$

$$= \frac{4.5}{E}$$

$$= \frac{-13}{E}$$

Vol. strain

$$\frac{\Delta V}{V} = \epsilon_x + \epsilon_y + \epsilon_z$$

$$\frac{\Delta V}{V} = \frac{11}{E} = 0.055 \times 10^{-3}$$

$$\Delta V = 0.055 \times (500 \times 100 \times 50)$$

$$= 137.5 \text{ mm}^3$$

Mohr circle

① A machine component is subjected to the stresses as shown in figure, find.

① Normal and shearing stress on section AB.

② maximum shear stress

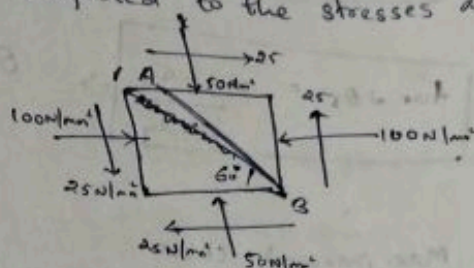
③ planes on which shear stress is maximum.

④ principal stresses.

⑤ principal planes

⑥ resultant stress on AB.

⑦ Resultant of maximum shear stress.



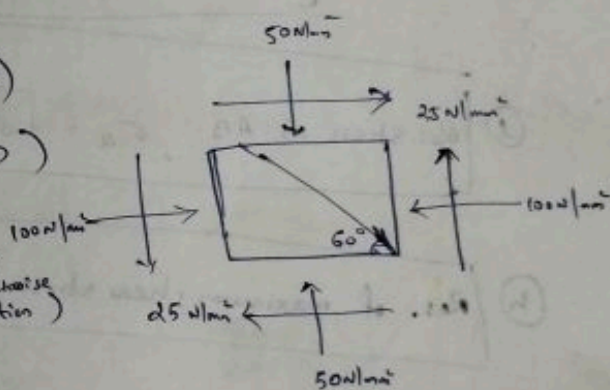
(Compression -ve)
(Tension +ve)
shear stress
C.W -ve
(Top & bottom)

$$\sigma_x = -100 \text{ (compression)}$$

$$\sigma_y = -50 \text{ (compression)}$$

$$\tau_{xy} = -25 \text{ (Top & bottom force in clockwise direction)}$$

$$\theta = 60^\circ$$



① Normal and shearing stress

$$\sigma_n = \frac{\sigma_x + \sigma_y}{2} - \left(\frac{\sigma_x - \sigma_y}{2} \right) \cos 2\theta - \tau_{xy} \sin 2\theta$$

$$= -65.85 \text{ N/mm}^2$$

$$\tau = \frac{\sigma_x - \sigma_y}{2} \sin 2\theta - \tau_{xy} \cos 2\theta$$

$$= -34.15 \text{ N/mm}^2$$

max. shear stress

$$\tau_{max} = \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \underline{\underline{35.25 \text{ N/mm}^2}}$$

$$\tan 2\theta_s = \frac{\sigma_x - \sigma_y}{2\tau_{xy}} \Rightarrow \theta_s = \underline{\underline{22.5^\circ}}$$

d) Max. principal stress

$$\sigma_{p_1} = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \underline{\underline{-39.64 \text{ N/mm}^2}}$$

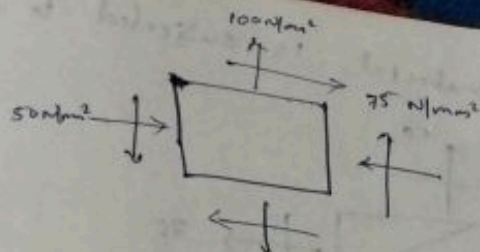
$$\sigma_{p_2} = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \underline{\underline{-110.35 \text{ N/mm}^2}}$$

e) principle plane.

$$\tan 2\theta_p = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} \Rightarrow \theta_p = \underline{\underline{22.5^\circ}}$$

f) Res. stress on AB, $\sigma_R = \sqrt{\sigma_n^2 + \tau^2} = \underline{\underline{74.17 \text{ N/mm}^2}}$

h) Res. of maximum shear stress = $\sqrt{\sigma_{p_1}^2 + \tau_{max}^2} = \underline{\underline{53.11 \text{ N/mm}^2}}$



- (a) Principal stress (b) max shear stress (c) plane of max shear stress

- (d) Resultant stress on plane of maximum shear stress.

$$\sigma_x = -50 \quad \sigma_y = 100 \quad \tau_{xy} = -75$$

$$(a) \sigma_{p_1} = \frac{\sigma_x + \sigma_y}{2} + \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \underline{\underline{131.06 \text{ N/mm}^2}}$$

$$\sigma_{p_2} = \frac{\sigma_x + \sigma_y}{2} - \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} = \underline{\underline{-81.06 \text{ N/mm}^2}}$$

- (b) max. shear stress

$$\tau_{max} = \underline{\underline{106.06 \text{ N/mm}^2}} \quad \left(\sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \right)$$

- (c) plane of max shear stress

$$\Rightarrow \tan 2\theta_s = \frac{\sigma_x - \sigma_y}{2\tau_{xy}}$$

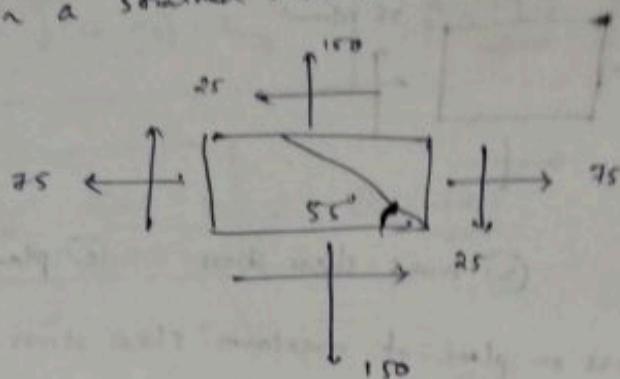
$$\theta_s = \frac{\tan^{-1} 1}{2} = \underline{\underline{22.5^\circ}}$$

- (d) Resultant stress on plane with max. shear stress.

$$R = \sqrt{\sigma_{p_1}^2 + \tau_{max}^2} = \sqrt{131.06^2 + 106.06^2}$$

$$= \underline{\underline{168.6 \text{ N/mm}^2}}$$

① A point in a strained material is subjected to the stresses as



Find τ_{max} , σ_{P_1} , σ_{P_2} , θ_P and σ_n , τ , σ_R , θ_s

$$\sigma_x = 75 \text{ N}$$

$$\sigma_y = 150$$

$$\tau_{xy} = 25$$

$$= 0 \text{ J} = 7.5 \text{ cm}$$

$$= 0 \text{ K} = 15 \text{ cm}$$

$$= 50 = 2.5 \text{ cm}$$

$$\tau_{max} = \text{radius}$$

$$\sigma_{P_1} = OG$$

$$\sigma_{P_2} = OH$$

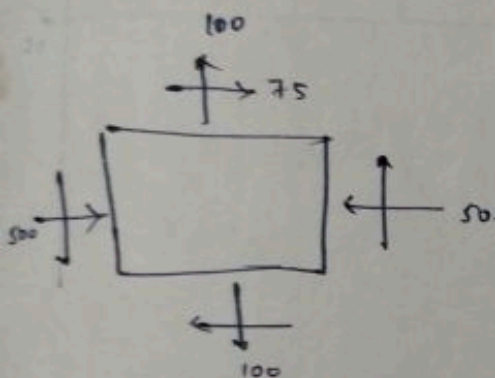
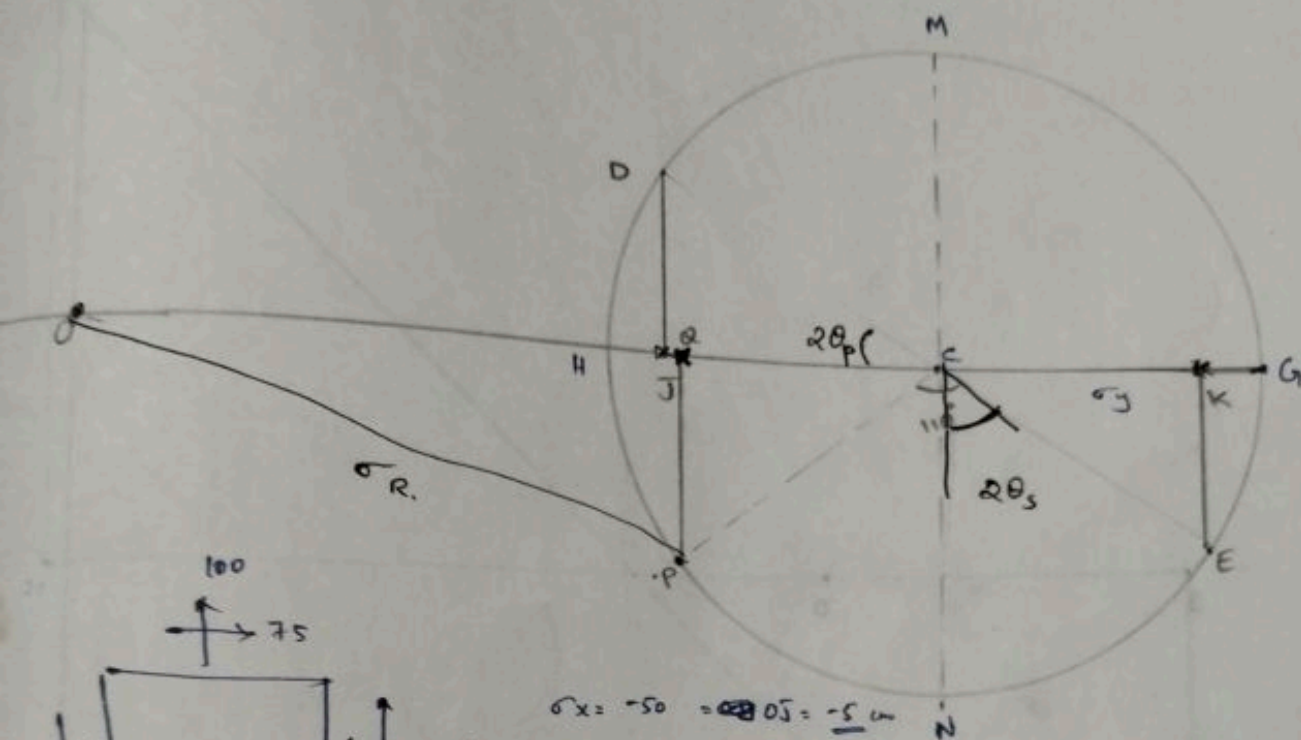
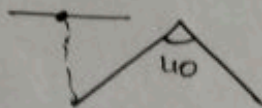
$$2\theta_P = \angle DCJ$$

$$\sigma_n = OQ$$

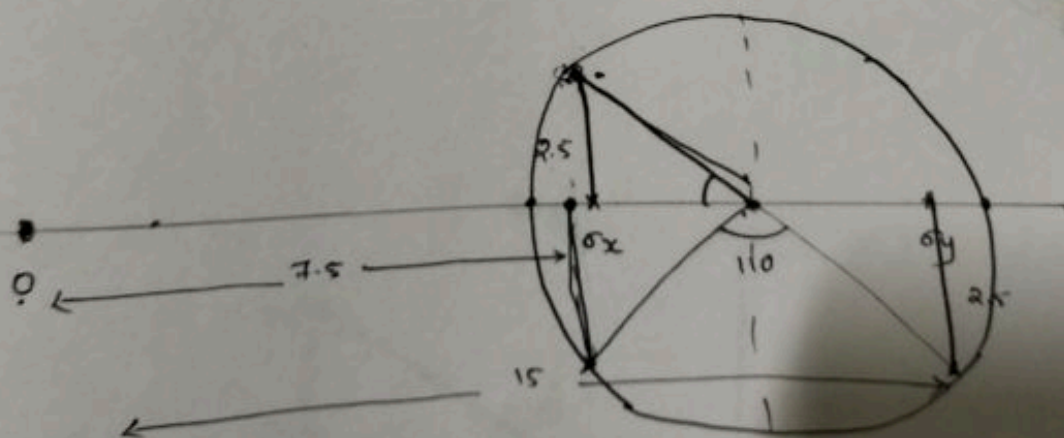
$$\tau = QP$$

$$\sigma_R = OP$$

$$2\theta_s = \angle ECN$$



$$\begin{aligned}\sigma_x &= -50 = -0.5 = -5 \text{ cm} \\ \sigma_y &= 100 = 10 \text{ cm} \\ \tau_{xy} &= -75 = -0.75 = -7.5 \text{ cm}\end{aligned}$$



Crackul. SS

Torsion

$$\frac{T}{J} = \frac{G\theta}{l} = \frac{\tau}{R}$$

* Solid shaft

Strength, $T = \frac{\pi}{16} \tau D^3 \text{ Nmm}$

$$J = \frac{\pi}{32} D^4 \text{ mm}^4$$

* Hollow shaft

Strength, $T = \frac{\pi}{16} \tau \left[\frac{D^4 - d^4}{D} \right] \text{ Nmm}$

$$J = \frac{\pi}{32} [D^4 - d^4] \text{ mm}^4$$

* Power of a shaft

$$P = \frac{2\pi NT}{60} \text{ (kW)}$$

* Polar modulus

$$Z_p = \frac{J}{R} \Rightarrow \text{mm}^3$$

T - Torque.

J = polar moment of inertia.

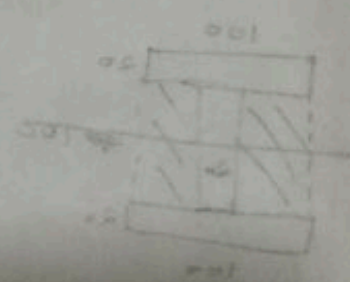
G = modulus of rigidity.

θ = angle of twist.

l = length of shaft.

τ = shear stress

R = radius



(2)

Q) A solid circular shaft of 100 mm diameter is transmitting 120 kW at 150 rpm. Find the intensity of shear stress in the shaft.

$$D = 100 \text{ mm}$$

$$P = 120 \text{ kW} = 120 \times 10^3$$

$$N = 150$$

$$T = \frac{\pi \tau D^3}{16} \text{ for solid shaft.}$$

$$\tau = \frac{16 T}{\pi D^3}$$

$$P = \frac{2 \pi N T}{60} \Rightarrow T = \frac{P \times 60}{2 \pi N}$$

$$= 7639.43 \text{ Nm}$$

$$= \frac{16 \times 7639.43 \times 10^3}{\pi \times 100^3} = \underline{\underline{38.9 \text{ N/mm}^2}}$$

$$T = \frac{7639.43 \times 10^3}{\text{Nm}}$$

Q) Find the angle of twist per metre length of a hollow shaft of 100 mm ext. and 60 mm internal diameters. If the shear stress is not to exceed 35 MPa. Take $G = 85 \text{ GPa}$.

$$D = 100, d = 60, \tau = 35, G = 85 \times 10^3$$

$$\frac{G \theta}{l} = \frac{T}{J} \Rightarrow \frac{\theta}{l} = \frac{T}{GJ}$$

$$T = \frac{\pi \tau (D^4 - d^4)}{16 D}$$

$$T = 5.98 \times 10^6 \text{ Nmm}$$

$$\frac{\theta}{l} = \frac{5.98 \times 10^6}{85 \times 10^3 \times 8.55 \times 10^6}$$

$$J = \frac{\pi (D^4 - d^4)}{32}$$

$$= 8.23 \times 10^{-6} \text{ rad/mm}$$

$$= 8.55 \times 10^6 \text{ mm}^4$$

$$= 8.23 \times 10^{-6} \times \frac{180 \times 10^3}{\pi} \text{ } ^\circ/\text{m}$$

$$= \underline{\underline{0.47^\circ/\text{m}}}$$

(3)

V.Q

Q) A solid Aluminium shaft 1 m long and 50 mm dia is to be replaced by a tubular shaft (steel) of same length and same outside diameter such that each of the two shafts could have the same angle of twist per unit torsional moment over the total length. What must be the inner diameter of the tubular steel shaft? Be 'G' of steel is 3 times that of Aluminium.

$$G_s = 3G_A, \quad l = 1000 \text{ mm} \quad D_A = 50 = D_s, \quad d_s = ?$$

$$l_A = l_s \quad \boxed{\frac{\theta_A}{T_A} = \frac{\theta_s}{T_s}} \quad \text{given}$$

$$\frac{T}{J} = \frac{G\theta}{l} \Rightarrow \boxed{\frac{\theta}{T} = \frac{l}{GJ}} \quad \frac{l_A}{G_A J_A} = \frac{l_s}{G_s J_s} = \frac{1}{G_A J_A} = \frac{1}{3G_A J_s}$$

$$\Rightarrow \frac{1}{J_{\text{Alum}}} = \frac{1}{3J_{\text{steel}}} \Rightarrow \frac{\pi D^4}{32} = \frac{1}{3} \frac{\pi (D^4 - d^4)}{32}$$

$$\Rightarrow 3(D^4 - d^4) = D^4$$

$$\Rightarrow 3D^4 - 3d^4 = D^4 \Rightarrow 2D^4 = 3d^4 \quad (D = 50)$$

$$\Rightarrow 2 \times 50^4 = 3d^4 \Rightarrow d = \sqrt[4]{\frac{2 \times 50^4}{3}} \quad \text{Ans}$$

(5)

Q.9

Q) A solid shaft of 200mm diameter has the same cross sectional area as that of a hollow shaft of same material with inside diameter of 150mm. Find the ratio of power transmitted by the two shafts at the same speed.

$$(\text{Solid Area}) \frac{\pi}{4} 200^2 = \frac{\pi}{4} (D^2 - 150^2) \quad (\text{hollow})$$

$$31415.9 = \frac{\pi}{4} (D^2 - 22500)$$

$$D = \underline{\underline{250 \text{ mm}}}$$

$$T_{\text{solid}} = \frac{\pi}{16} \tau_1 200^3 = 15.7 \times 10^5 \tau_1$$

$$T_{\text{hollow}} = \frac{\pi}{16} \tau_2 \left(\frac{250^4 - 150^4}{250} \right) = 26.7 \times 10^5 \tau_2$$

$$\frac{P_1}{P_2} = \frac{T_1}{T_2} \quad N_1 = N_2$$

$$\frac{2\pi N T_1}{60} = \frac{2\pi N T_2}{60}$$

$$\Rightarrow 15.7 \times 10^5 \tau_1 = 26.7 \times 10^5 \tau_2$$

$$\frac{P_1}{P_2} = \frac{15.7 \times 10^5 \tau_1}{26.7 \times 10^5 \tau_2}$$

$$= 0.58$$

$$\underline{\underline{P_1 = 0.58 P_2}}$$

assuming

$\tau_1 = \tau_2$ since
material is same

Q.2

(4)

a) A solid shaft is to transmit 500kW power at 170rpm. The shear stress in the material is not exceed 80mpa. Shaft is replaced by a hollow shaft $[d=0.6D]$. Find the diameter required. Maximum shear stress being same (80mpa). Calculate the Percentage saving in material.

Solid shaft

$\rightarrow P = 500 \text{ kW}, N = 170 \text{ rpm}, \tau = 80 \text{ MPa}$

$D = ?$

$$T = \frac{P \times 60}{2\pi N} = \frac{500 \times 10^3 \times 60}{2\pi \times 170} = 28086.16 \text{ Nmm} \times 10^3$$

~~$$T = \frac{\pi}{32} D^4$$~~

$$T = \frac{\pi \tau D^3}{16}$$

$$\rightarrow D^3 = \frac{T \times 16}{\pi \tau}$$

$$D = \sqrt[3]{\frac{28086.16 \times 10^3 \times 16}{\pi \times 80}} = 121.37 \text{ mm}$$

Hollow shaft

$$T = \frac{\pi \tau (D^4 - d^4)}{16 D}$$

~~$$T = \frac{\pi \tau (D^4 - (0.6D)^4)}{16 D}$$~~

~~$$= \frac{\pi \times 80 (121.37^4 - (0.6 \times 121.37)^4)}{16 \times 121.37}$$~~

Percentage saving in material

$$\frac{A_1 - A_2}{A_1} = \frac{\frac{\pi}{4} D^2 - \frac{\pi}{4} (D^2 - d^2)}{\frac{\pi}{4} D^2}$$

$$\approx \underline{\underline{30\%}}$$

$$T = \frac{\pi \tau (D^4 - (0.6D)^4)}{16 D}$$

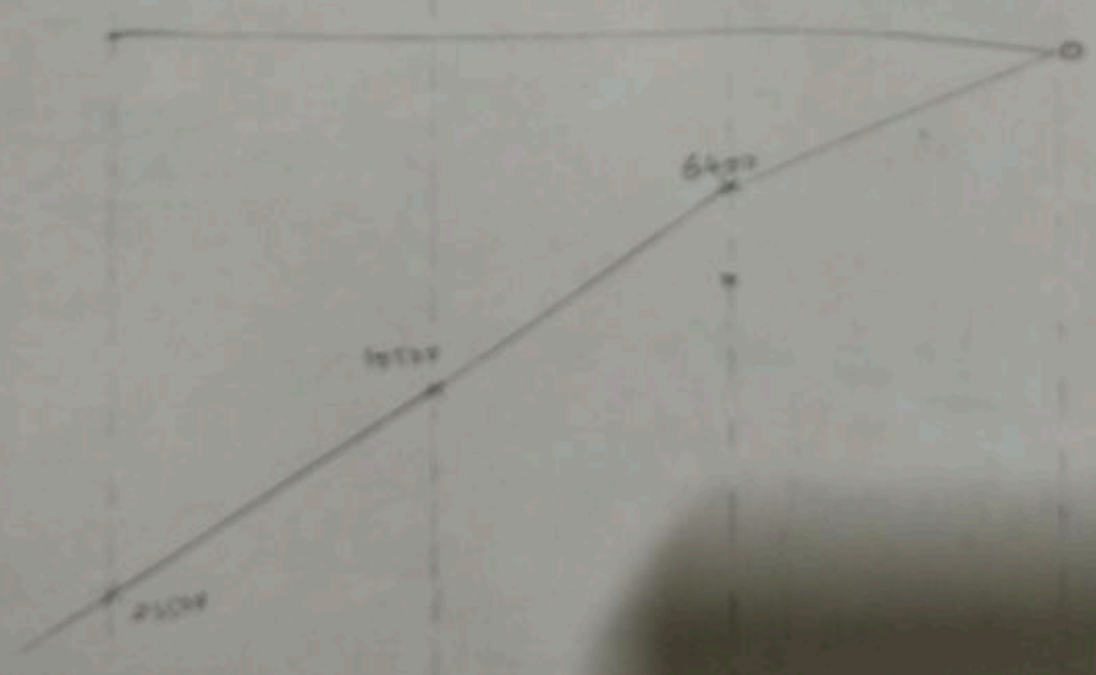
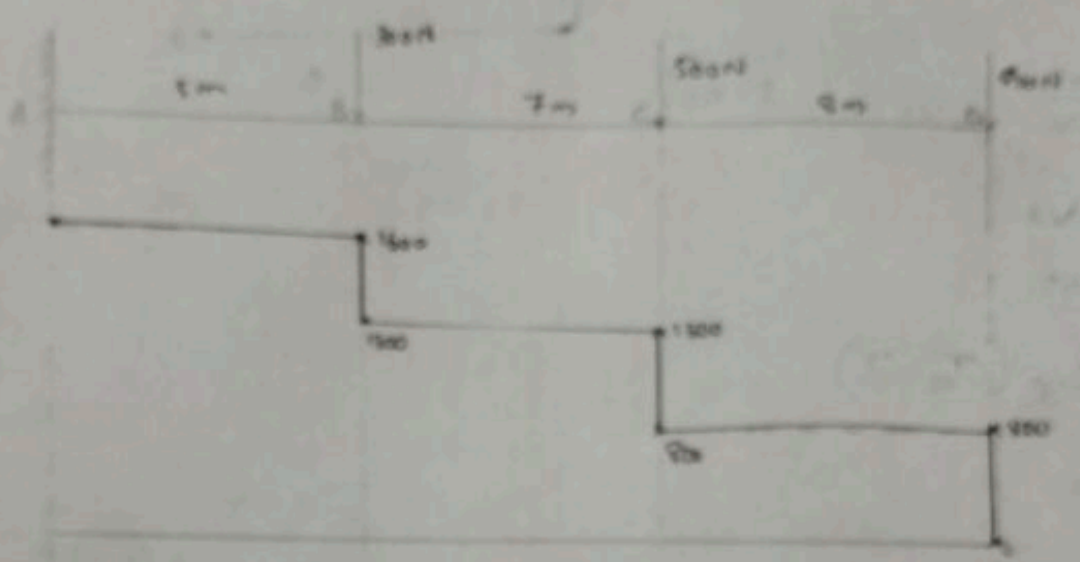
$$\frac{T \times 16}{\pi \tau} = \frac{D^4 - 0.1296 D^4}{D} = 0.8704 D^3$$

$$\Rightarrow D = \sqrt[3]{\frac{28086.16 \times 10^3 \times 16 \times 0.8704}{\pi \times 80}}$$

$$= 127.12 \text{ mm}$$

$$d = 0.6D = 76.2 \text{ mm}$$

$50 \times 7 = 350$
 $400 \times 7 = 2800$
 $2800 + 350 = 3150$
 $3150 \div 10 = 315$



550

$$D \rightarrow 0 \text{ M}$$

$$O \rightarrow -800 \text{ M}$$

$$C \rightarrow -800$$

$$C \rightarrow -800 - 500 = -1300 \text{ M}$$

$$B \rightarrow -800 - 500 = -1300 \text{ M}$$

$$B \rightarrow -800 - 500 - 300 = -1600 \text{ M}$$

$$A \rightarrow -800 - 500 - 300 = -1600 \text{ M}$$

67

$$D = 0$$

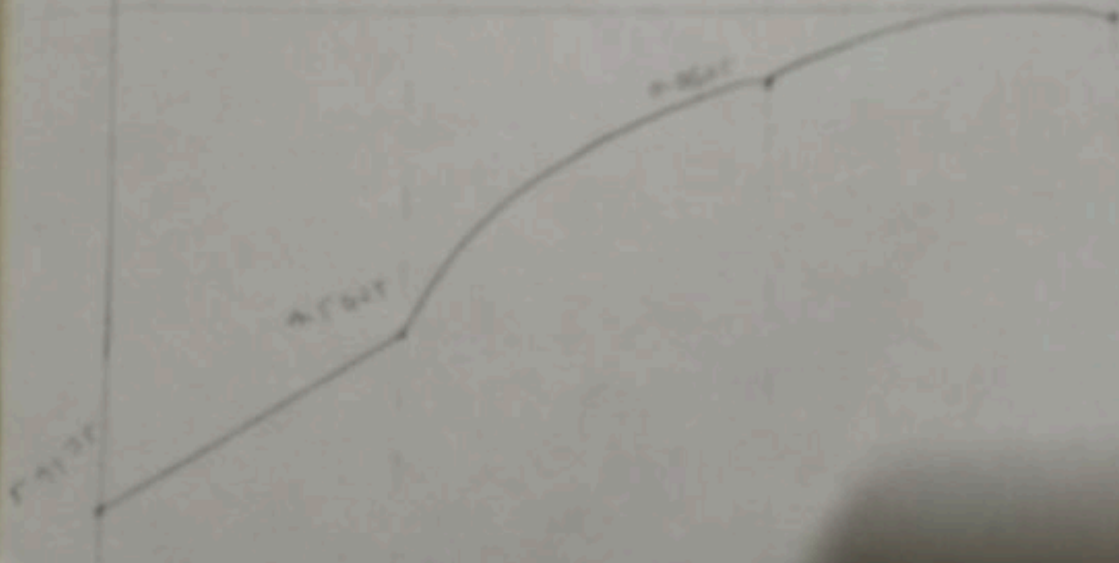
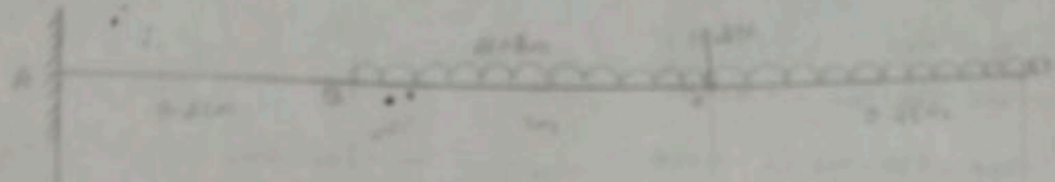
$$C = -800 \times 2 = -1600$$

$$B = -800 \times 15 + -500 \times 7 = -15500$$

$$A = -800 \times 20 + -500 \times 12 + -300 \times 5 = -21500$$

UDL

(4000 N/m)



$U = U_E - U_D - \dots$
 UOL - magnitude distance points multiply by sign (SFD)

SFD $W \uparrow = C \uparrow + D \uparrow$

$D = 0$

$C = -2 \times 0.25 = -0.5$

$C = -2 \times 0.25 - 3 = -3.5$

$B = -2 \times 1.25 - 3 = -5.5$

$A = -2 \times 1.25 - 3 = -5.5$

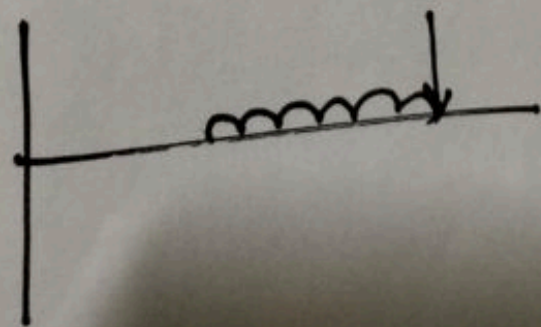
BM

$M_D = 0$

$M_C = -2 \times 0.25 \times \frac{0.25}{2} = -0.0625$

$M_B = -2 \times 1.25 \times \frac{1.25}{2} - 3 \times 1 = -4.5625$

$M_A = \left[-2 \times 1.25 \times \left(\frac{1.25}{2} + 0.25 \right) \right] - 3 \times 1.25 = -5.9375$



Simply supported

D.M.

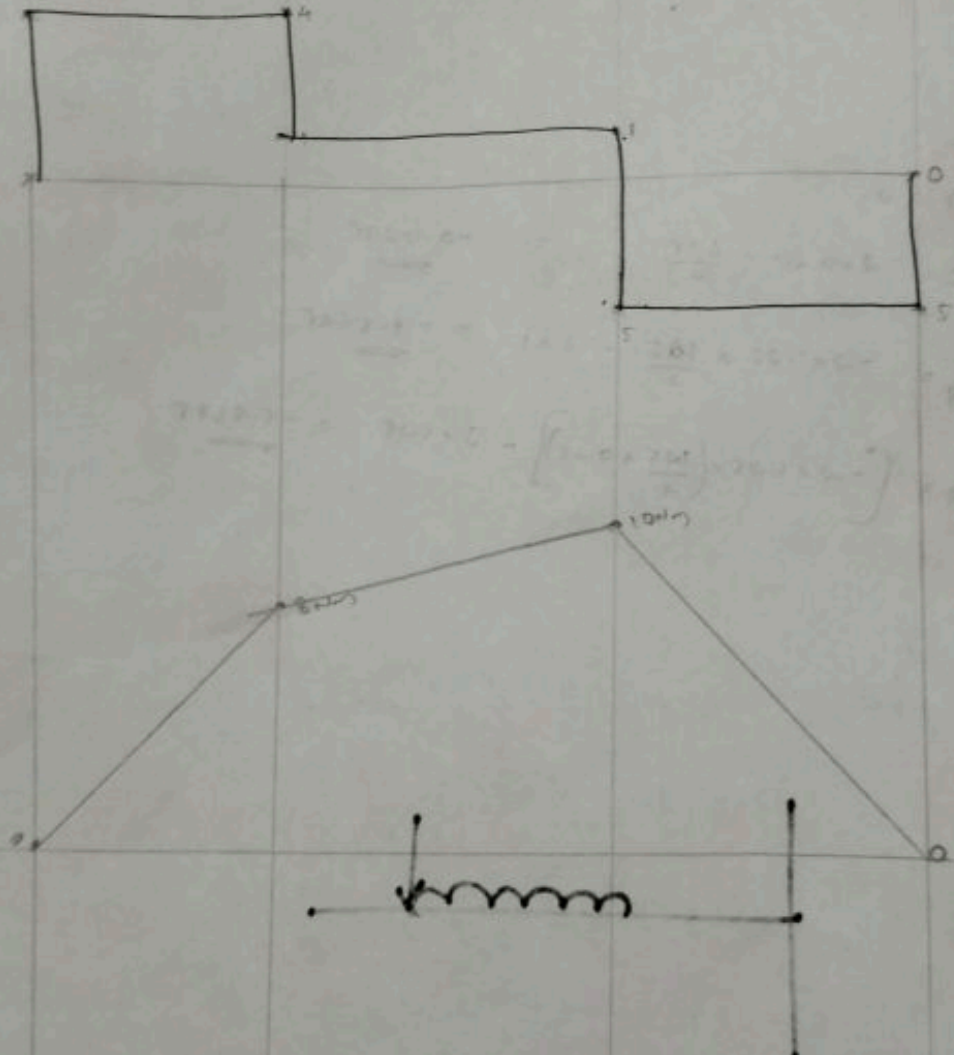
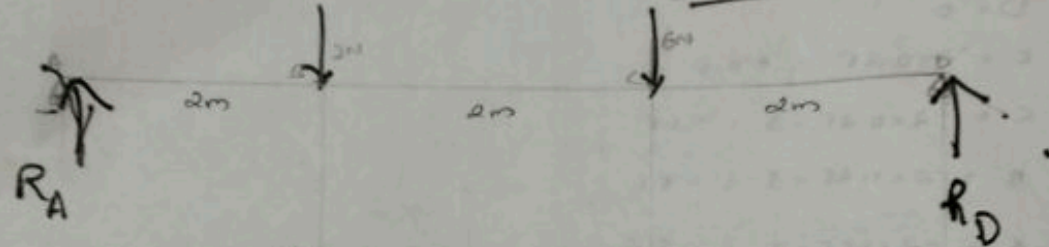
Reaction

Clockwise -ve

Anticlockwise +ve

$$R_A + R_D - 6N - 3N = 0$$

$$\underline{\underline{R_A + R_D = 9N}}$$





force $\rightarrow R_A - 3 - 6 + R_D = 0$

moment $\rightarrow \sum M_A = -3 \times 2 - 6 \times 4 + R_D \times 6 = 0$ $R_A + R_D = 9$

$R_D = 5, R_A = 4$

SFD

$D = 0$

$D = 5$

$C = 5$

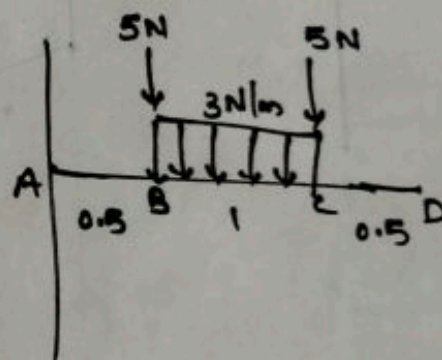
$C = -6 + 5 = -1$

$B = -6 + 5 = -1$

$B = -3 - 6 + 5 = -4$

$A = 5 - 3 - 6 = -4$

$A = 5 + 4 - 3 - 6 = 0$



BM

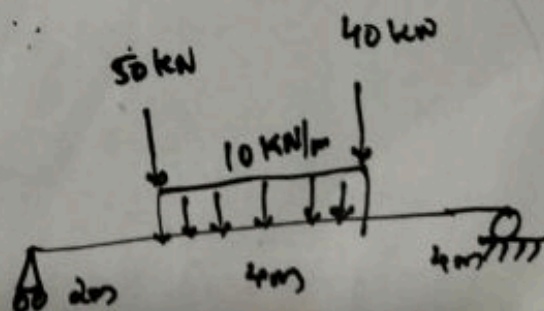
$M_D = 0$

$M_C = 5 \times 2 = 10$

$M_B = 5 \times 4 - 6 \times 2 = 8$

$M_B = 5 \times 4 - 6 \times 2 = 0$

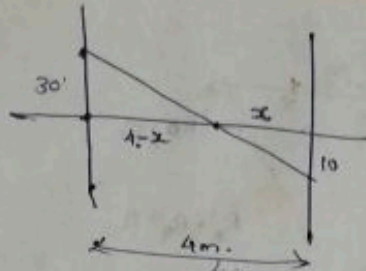
$M_A = 5 \times 6 - 6 \times 4 - 3 \times 2 = 0$



40x

$$\frac{4-x}{30} = \frac{x}{10}$$

$$x = 1$$



$$\frac{10}{x} = \frac{30}{4-x}$$

$$40 - 10x = 30x$$

$$40 = 40x$$

$$x = 1$$

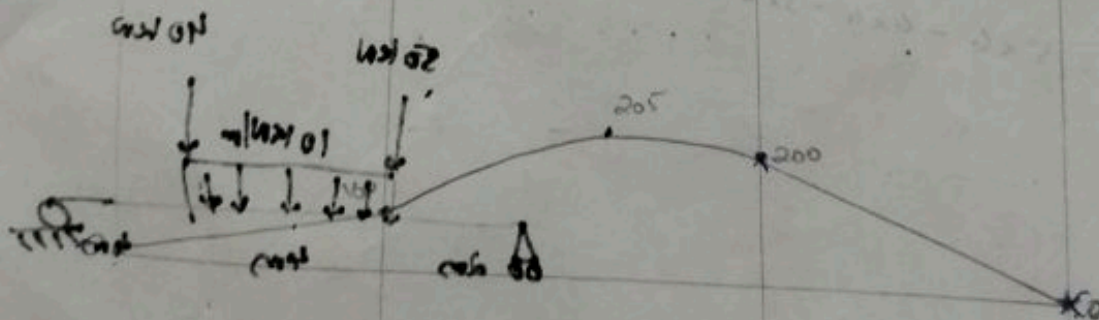
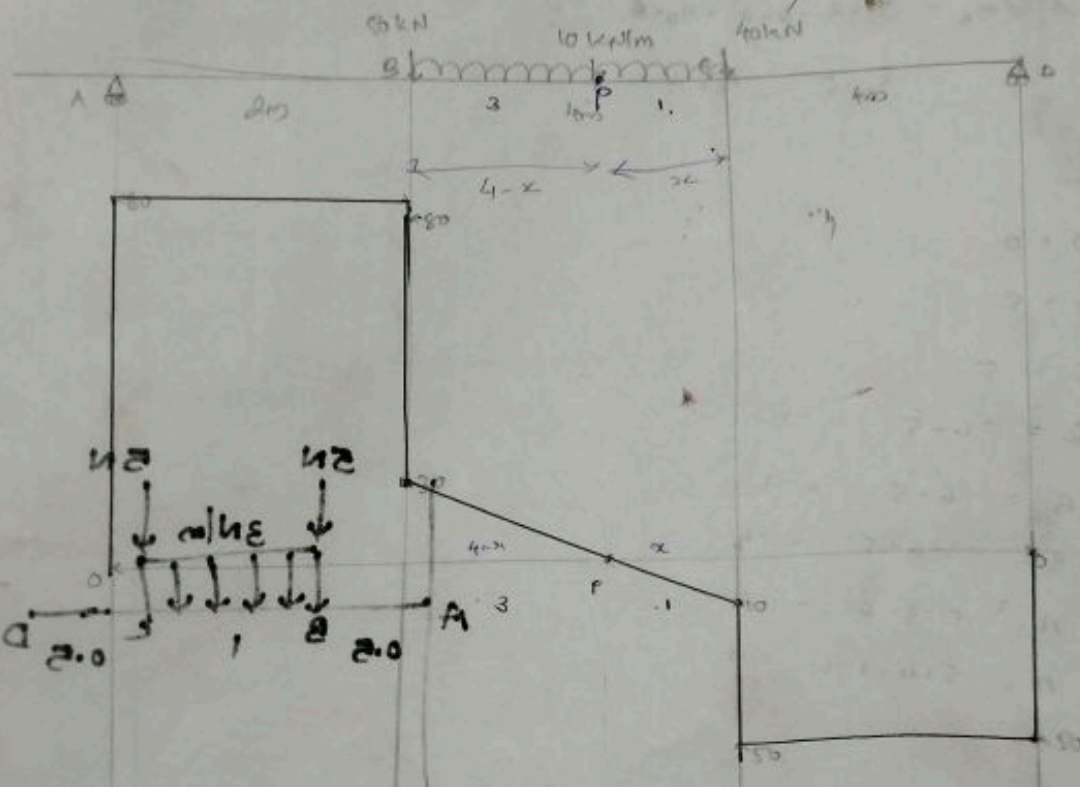


Diagram of a beam with a reaction R_A at the left end, a uniformly distributed load of 10 kN/m over a 4 m span, and a point load of 40 kN at the end of the span. The total length of the beam is 10 m , with a reaction R_D at the right end.

BM of UDL $\Rightarrow (F \times d) \left(\frac{d}{2} + x \right)$
 $= (10 \times 4) \left(\frac{4}{2} + 2 \right) = 40 \times 4$

$$R_A - 50 - (10 \times 4) - 40 + R_D = 0$$

$$\sum M_A = -50 \times 2 - 40 \times 4 - 40 \times 6 + R_D \times 10$$

$$R_D = 50, R_A = 80$$

SFD

$D = 0$

$D = 50$

$C = 50$

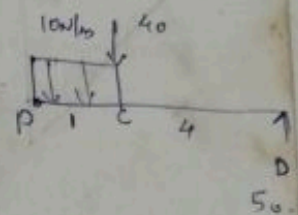
$C = -40 + 50 = 10$

$B = (-10 \times 4) - 40 + 50 = -30$

$B = -50 - (10 \times 4) - 40 + 50 = -80$

$A = 0$

$A = 80 - 50 - (10 \times 4) - 40 + 50 = 0$



$M_P = 50 \times 5 - 40 \times 1$

$-10 \times 1 \times \frac{1}{2}$

$= 250 - 40 - 5$

$= 205$

BM

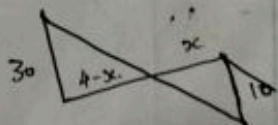
$M_D = 0$

$M_C = 50 \times 4 = 200$

$M_B = 50 \times 8 - 40 \times 4 - \left[10 \times 4 \times \left(\frac{4}{2} \right) \right] = 160$

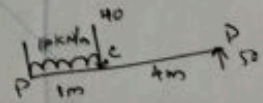
$M_A = 50 \times 10 - 40 \times 6 - \left[10 \times 4 \times \left(\frac{4}{2} + 2 \right) \right] - 50 \times 2 = 0$

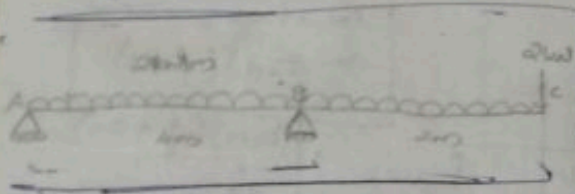
Point of Contraflexure (P)



$M_P = 50 \times 5 - 40 \times 1 - 10 \times 1 \left(\frac{1}{2} \right)$
 $= 205$

$\frac{40 - x}{30} = \frac{x}{10} \Rightarrow x = 1$





$$R_A - (2 \times 6) + R_B - 2 = 0$$

$$R_A + R_B = 14$$

$$\sum M_A = -2 \times 6 + R_B \times 4 - (2 \times 6 \times \left[\frac{6}{2}\right]) = 0$$

$$-12 + 4R_B - 36 = 0$$

$$R_B = 12, R_A = 2$$

SFD

$$C = 0$$

$$C = -2$$

$$B = -2 \times 4 - (-2 \times 4) = -6$$

$$B = 12 - 2 - (2 \times 2) = 6$$

$$A = -2 - (2 \times 6) + 12 = -2$$

$$A = -2(-2 \times 6) + 12 + 2 = 0$$

BM

$$M_C = 0$$

$$M_B = -2 \times 2 - (2 \times 2 \left[\frac{2}{2}\right]) = -8$$

$$M_A = -2 \times 6 - (2 \times 6 \left[\frac{6}{2}\right]) + 12 \times 4 = 0$$

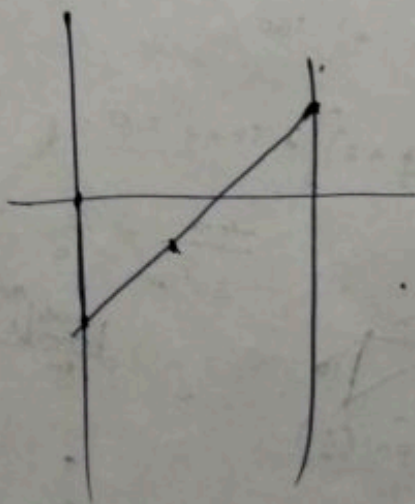
Point of contraflexure (P)

$$\frac{x}{6} = \frac{4-x}{2} \Rightarrow x = 3$$



$$M_P = -2 \times 5 + 12 \times 3 - (2 \times 5 \times \frac{5}{2})$$

$$= 1$$





$R_A, R_B = 0.10$

$$R_B = 49.16, R_D = 60.84$$

540

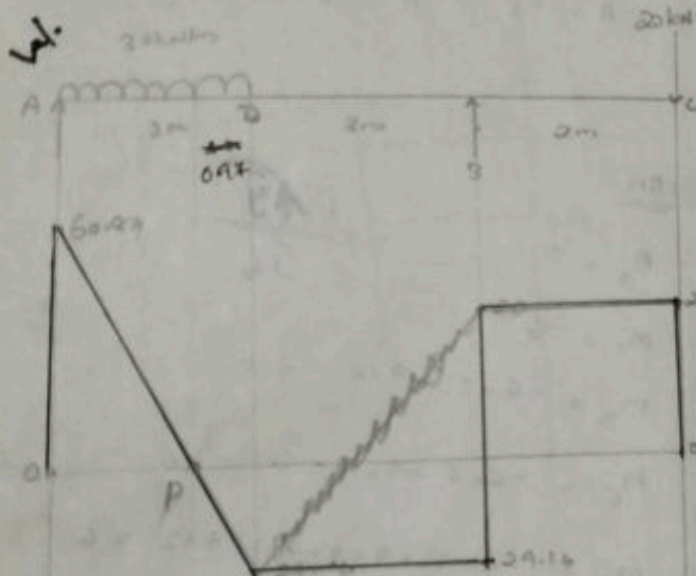
$$c = 0, \quad c = -20$$

$$B = -20, \quad B = 29.14$$

$D = 29.6$ 

$$A = -20 + 49.16 - 30 \times 3$$
$$= -60.84$$

$$A = 0$$

BM

$$GM_t = 0$$

$$M_B = -20 \times 2 = -40$$

$$M_D = -20 \times 5 + 49.16 \times 3$$

$$= 47.48 \approx \underline{\underline{47.5}}$$

$$M_A = -20 \times 8 + 49.16 \times 6 - 30 \times 3 \times \frac{3}{2}$$

Point of contraflexure. (P)

$$\frac{x}{29.16} = \frac{3-x}{60.84}$$

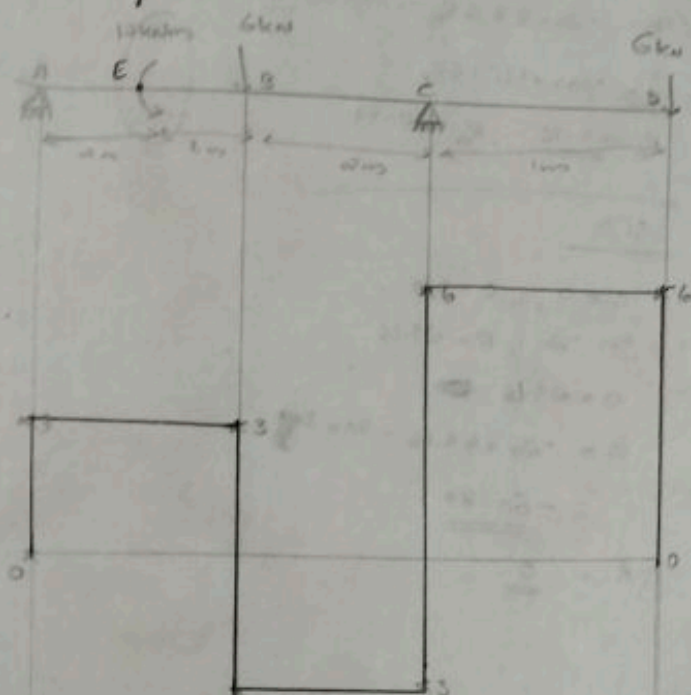
$$60.84x = 87.48 - 29.16x$$

$\gamma = 0.47$

$$M_p = -20 \times \frac{5.97}{2} + 49.16 \times \frac{3.97}{2}$$

$$= -155.12 + 1283.16 - 184.21$$

(just add it on calculating moment)



$$R_A - 6 + R_C - 6 = 0 \Rightarrow R_A + R_C = 12$$

$$\sum M_A = (-6 \times 4) + R_C \times 6 + (-6 \times 7) + 12 = 0$$

$$\Rightarrow R_C = 9, R_A = 3$$

QFD

$$D = 0, D = -6$$

$$C = -6, C = -6 + 9 = 3$$

$$B = 3, B = -6 + 9 - 6 = -3$$

$$A = -3, A = -6 + 9 - 6 + 3 = 0$$

BM

$$M_D = 0$$

$$M_C = -6$$

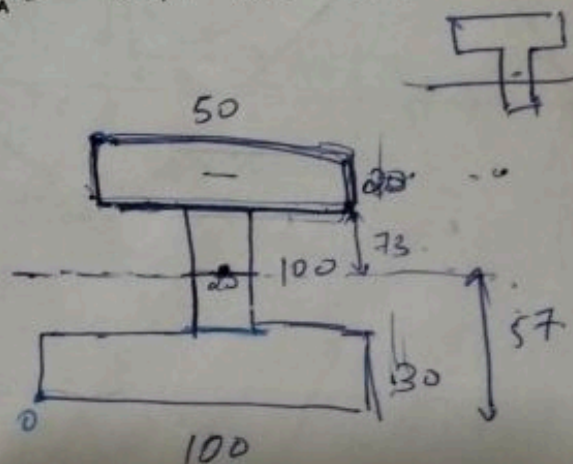
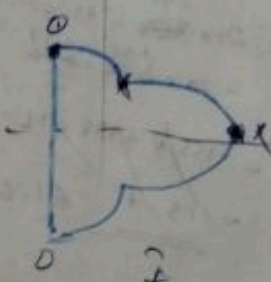
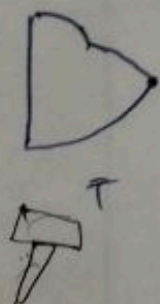
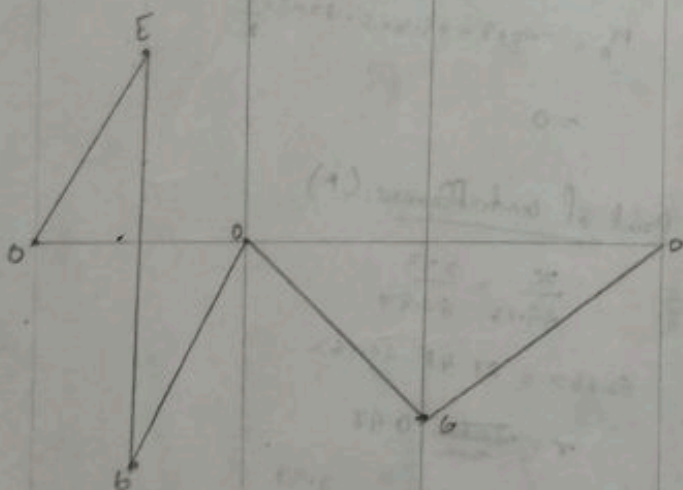
$$M_B = -6 \times 3 + 9 \times 2 = 0$$

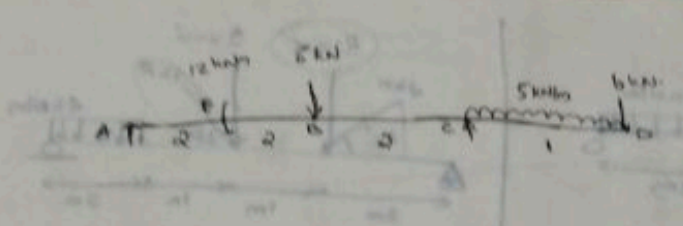
$$M_E = -6 \times 5 + 9 \times 4 - 6 \times 2 = -6$$

$$M_E = -6 \times 5 + 9 \times 4 - 6 \times 2 + 12 = 6$$

(moment direct angle) always case in mm
also add on zero).

$$M_A = -6 \times 7 + 9 \times 6 - 6 \times 4 + 12 = 0$$





$$R_A + R_C = 6 + 6 = 12$$

$$R_A + R_C = 12$$

$$\sum M_A = -6 \times 2 - 5 \times (2 \times 2) - 6 \times 4 + 12 \times 6 = 0$$

$$= -12 - 20 - 24 + 72 = 16$$

$$= R_C = 16.4$$

$$R_A = 2.6$$

SFO

$$D = 0, D = -6$$

$$C = -6 - 5 \times 1 = -11, C = 3.4$$

$$B = 3.4, B = 9 - 2.6$$

$$A = -2.6, A = 0$$

BM

$$D = 0$$

$$C = -6 - 5 \times 1 \times \frac{1}{2} = -8.5$$

$$B = -6 \times 3 + 5 \times (\frac{1}{2} \times 2) + 12 \times 2$$

$$= -1.7$$

$$M_E = -(-6 \times 5) + (5 \times (\frac{1}{2} \times 4))$$

$$+ 12 \times 4 - 6 \times 2$$

$$= -6.9$$

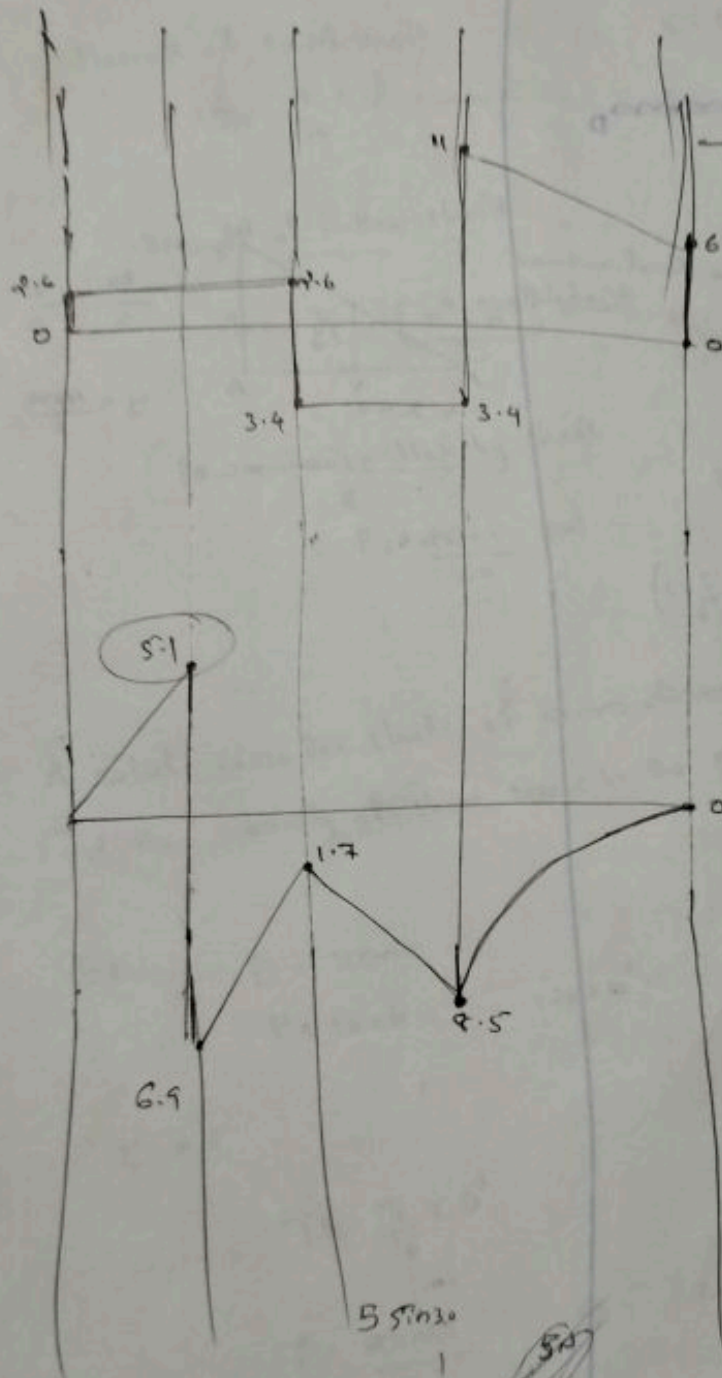
$$M_G = -12 + 12 = 0$$

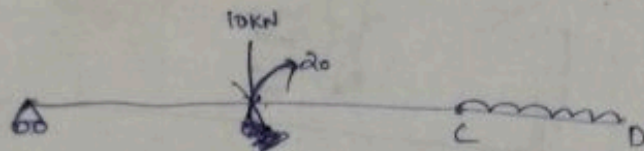
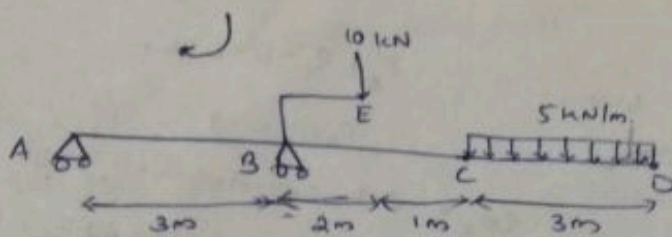
$$= 5.1$$

$$M_A = -6 \times 2 - 5 \times (1 \times 2)$$

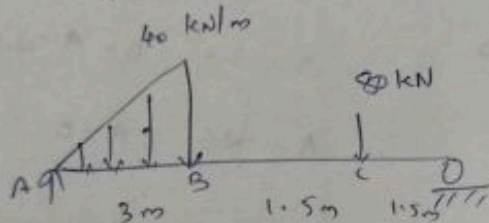
$$+ 12 \times 6 - 6 \times 4$$

$$+ 12 = -0.1$$





Centroid is $\frac{1}{3}$ from base and the focal will move $\frac{2}{3}$ of distance

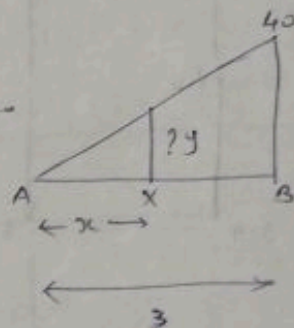
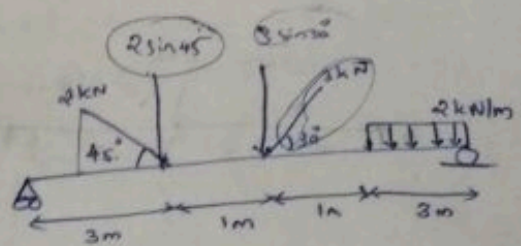


$$\sum M_p = 6RD - 80 \times 4.5 - \frac{1}{2} \times 3 \times 40 \left(\frac{2}{3} \times 3 \right)$$

$$6RD = 480 \Rightarrow RD = 80 \text{ kN}$$

$$R_A + R_D = \frac{1}{2} \times 3 \times 40 + 80 = 140$$

$$R_A = \underline{\underline{60 \text{ kN}}}$$



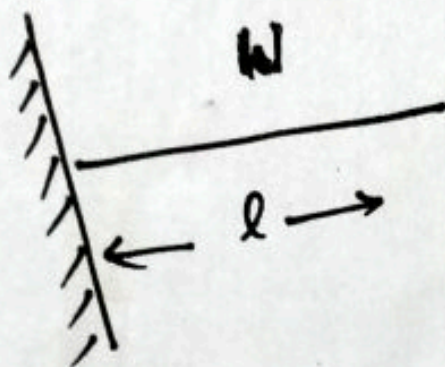
$$\frac{40}{3} = \frac{y}{x}$$

$$y = \frac{400x}{3}$$

③

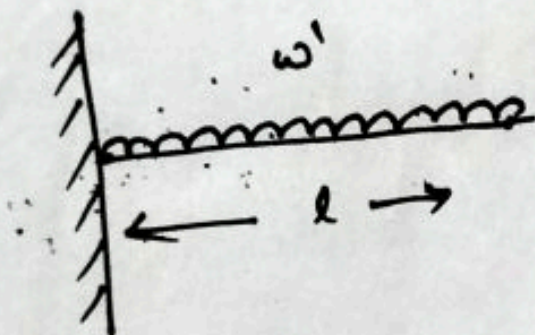
Stress | Torsion

Stress $\left\{ \begin{array}{l} \rightarrow \text{Normal / Bending stress} \\ \rightarrow \text{Torsional / Shear stress} \end{array} \right.$



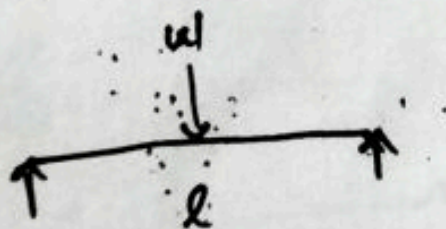
$$M = Wl$$

$$F = W$$



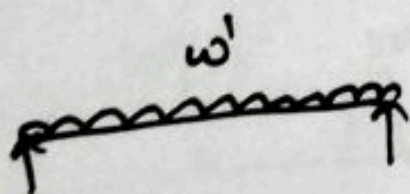
$$M = \frac{w'l^2}{2}$$

$$F = w'l$$



$$M = \frac{wl}{4}$$

$$F = \frac{w}{2}$$



$$M = \frac{w'l^2}{8}$$

$$F = \frac{w'l}{2}$$

Thin walled pressure vessels

For thin cylindrical vessel

$$\sigma_1 \text{ (Circumferential stress) } = \frac{Pr}{t} = \frac{Pd}{2t} \quad \begin{matrix} p = \text{Internal Pressure} \\ t = \text{wall thickness} \end{matrix}$$

| hoop stress

$$\sigma_2 \text{ (Longitudinal stress)} = \frac{Pr}{2t} = \frac{Pd}{4t} = \underline{\underline{\frac{\sigma_1}{2}}}$$

For thin spherical vessel

$$\sigma_1 = \sigma_2 = \frac{Pd}{4t}$$

affixing

$$\sigma = \frac{Pd}{4t}$$

(Only circumferential stress)

$$\text{Volumetric strain} = \frac{\Delta V}{V} = 2 \text{ Cir. strain} + \text{Long. strain}$$

$$= \frac{Pd}{2tE} \left(\frac{\sigma_1}{2} - \frac{2}{3} \right) \quad \left[\frac{1}{m} = \mu \right]$$

$$\text{Vol. stress} = P$$

$$\text{Bulk modulus, } K = \frac{\text{Vol. stress}}{\text{Vol. strain}}$$

$$\text{Circumferential strain, } \epsilon_1 = \frac{Pd}{2tE} \left(1 - \frac{1}{2m} \right)$$

$$\text{Longitudinal strain, } \epsilon_2 = \frac{Pd}{2tE} \left(\frac{1}{2} - \frac{1}{m} \right)$$

Volumen of cylinder

$$V = \frac{\pi}{4} d^2 l$$

- Q) A thin cylinder shell of internal diameter 2m and thickness 10mm is subjected to internal fluid pressure 4Mpa. Calculate longitudinal stress, hoop stress, Vol. strain
(Take $G = 200 \text{ GPa}$, Poisson's ratio = 0.3)

Failure of columns

$$P_E = \frac{\pi^2 EI}{L_e^2}$$

* Both ends hinged

$$L_e = l$$

* One end fixed & other free

$$L_e = 2l$$

* Both ends fixed

$$L_e = l/2$$

* One end fixed and other end hinged

$$L_e = l/\sqrt{2}$$

$$\text{Safe load} = \frac{\text{Crippling load}}{\text{FOS}}$$

$$\text{Slenderness ratio} = \frac{\text{Actual length}}{\text{Radius of gyration}} = \frac{l}{k}$$

$$k = \sqrt{\frac{I}{A}} \quad \begin{matrix} \text{moment of inertia} \\ \text{area} \end{matrix}$$

P_E = crippling load

L_e = Effective length

EI = flexural rigidity

$$I_{\text{round}} = \frac{\pi d^4}{64}$$

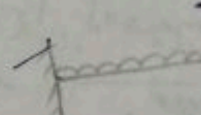
$$I_{\text{hollow}} = \frac{\pi (O^4 - d^4)}{64}$$

$$I_{\text{bar real}} = \frac{bd^3}{12}$$

$$I_{\text{square}} = \frac{a^4}{12}$$



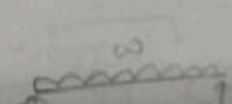
$$y = \frac{Wl^3}{3EI} \quad i = \frac{Wl^2}{2EI}$$



$$y = \frac{wl^4}{8EI} \quad i = \frac{wl^3}{6EI}$$



$$y = \frac{Wl^3}{48EI} \quad i = \frac{Wl^2}{16EI}$$



$$y = \frac{5wl^4}{384EI} \quad i = \frac{wl^3}{24EI}$$

3) Q) A column of 10m long has a section $300\text{mm} \times 500\text{mm}$. Determine the Euler's crippling load if

- i) both ends of column are fixed.
- ii) both ends of column are pinned.

1) Q) For solid round bar 3m long & 5cm in dia is used as a strut ($E = 2.0 \times 10^5 \text{ N/mm}^2$). Determine the crippling load, when the given strut is used with following condition.

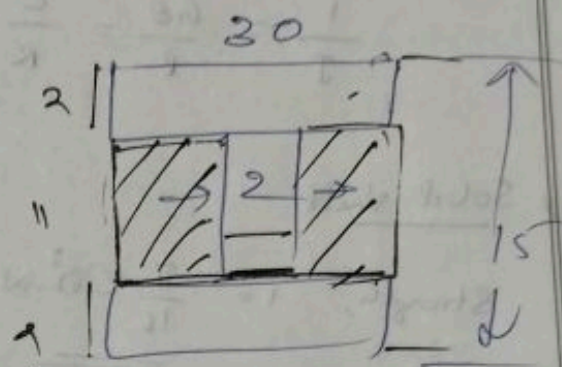
- i) one end fixed & other end free.
- ii) one end fixed & other is hinged.



2) A column of timber section $15\text{cm} \times 20\text{cm}$ is 6m long both ends being fixed. If the young's modulus of timber $= 17.5 \text{ kN/mm}^2$ determine the crippling load, safe load if FOS is 3. ($\times 10^3$)

4) A simply supported beam of length 4m is subjected to UDL of 30 kN/m over whole span and deflects 15mm at centre. Determine the crippling load if both ends of column is pinned.

- Q) Determine Euler's crippling load for an I section joist 30 cm x 15 cm and 5 m long which is used as strut with both ends fixed. Take Young's modulus for the joist $2 \times 10^5 \text{ N/mm}^2$.

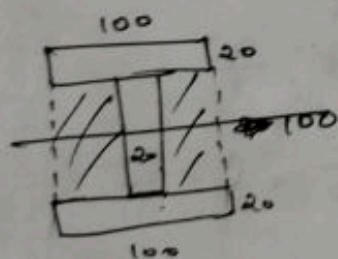


$$I = \frac{bd^3}{12} + A_1(\bar{y}-y_1)^2$$

$$+ \frac{bd^3}{12} + A_2(\bar{y}-y_2)^2$$

$$+ \frac{bd^3}{12} + A_3(\bar{y}-y_3)^2$$

$$\bar{y} = \frac{A_1 y_1 + A_2 y_2 + A_3 y_3}{A_1 + A_2 + A_3}$$



$$I = \frac{100 \times 140^3}{12} - \frac{20 \times 100^3}{12}$$

$$= \underline{\underline{\quad \quad \quad}}$$



Symmetric

$\bar{y}$ = centre.

Deflection of beams

Step ① : Find Moments of all sections

Step ② : Write Macaulay's Equation.

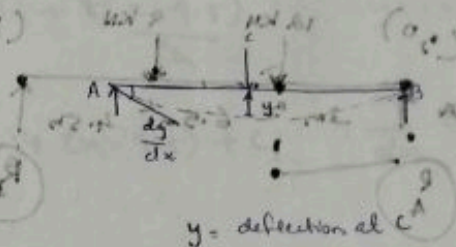
Step ③ : Apply ~~boundary~~ ^{integration} conditions to get slope equation

Step ④ : Take second derivative integral to get equation of deflection.

Step ⑤ : Apply boundary conditions to get C_1 & C_2

3000 N

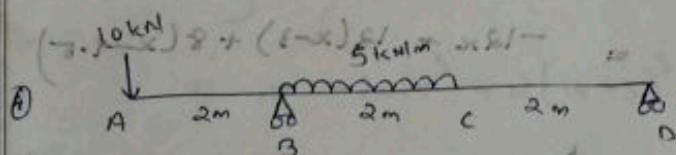
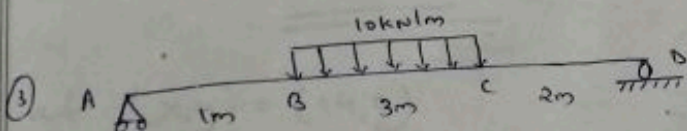
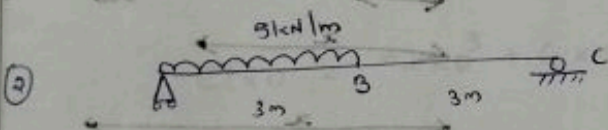
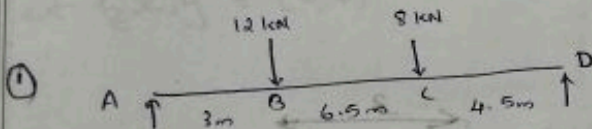
3000 N



$\frac{dy}{dx}$ at $(0,0) = \text{slope at } A$

* Maximum deflection occurs where slope is 0.

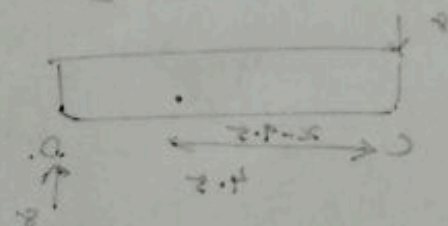
$\frac{d^2y}{dx^2} = \frac{w}{EI}$



$(7.5 \times x) + (5 \times x) = \frac{w \cdot x^2}{2}$



$(P - x) \times x = \frac{w \cdot x^2}{2}$



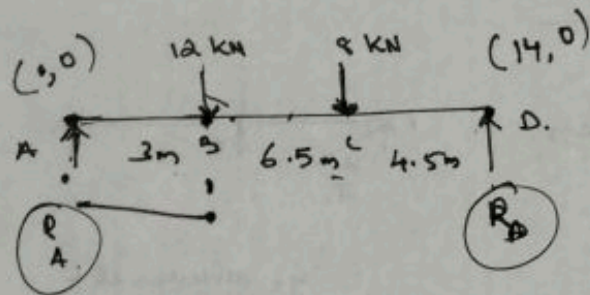
$(P - x) \times x = \frac{w \cdot x^2}{2}$

$\frac{d^2y}{dx^2} = \frac{w}{EI}$

$\frac{d^2y}{dx^2} = \frac{w}{EI}$

moment
C.W $\Rightarrow$ -ve
C.C.W $\Rightarrow$ +ve

Eqn



$$R_A + R_D = 12 + 8$$

$$= \underline{\underline{20 \text{ kN}}} \quad \text{--- (1)}$$

$$M_A = R_D \times 14 - 8 \times 9.5$$

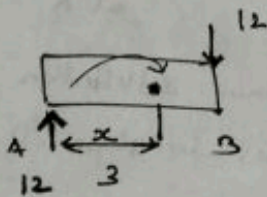
$$- 12 \times 3 = 0$$

$$= 14R_D - 76 - 36$$

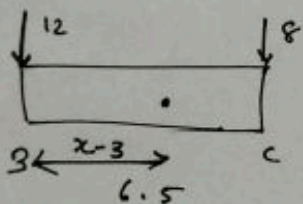
$$14R_D = 112$$

$$R_D = \underline{\underline{8 \text{ kN}}}$$

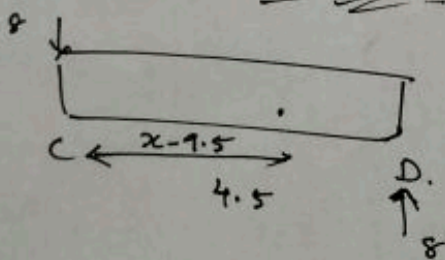
$$R_A = \underline{\underline{12 \text{ kN}}}$$



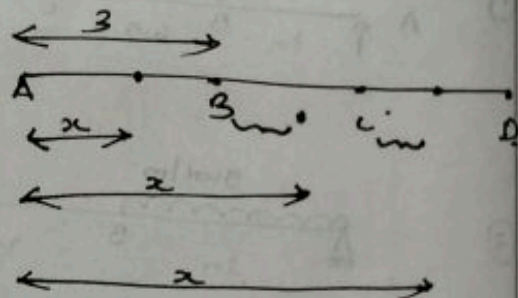
$$M_1 = \underline{\underline{-12x}}$$



$$M_2 = \underline{\underline{12(x-3)}}$$



$$M_3 = \underline{\underline{8(x-9.5)}}$$



$$M = M_1 + M_2 + M_3 = -12x + 12(x-3) + 8(x-9.5)$$

$$\frac{EI d^2 y}{dx^2} = M$$

$$\Rightarrow \frac{EI d^2 y}{dx^2} = -12x + 12(x-3) + 8(x-9.5)$$

$$EI \frac{dy}{dx} = -\frac{6}{1}x^2 + \frac{6}{2}(x-3)^2 + \frac{4}{3}(x-9.5)^3 + C_1$$

Slope

$$EI \frac{dy}{dx} = -6x^2 + 6(x-3)^2 + \frac{4}{3}(x-9.5)^3 + C_1$$

$$EI y = -\frac{6}{3}x^3 + \frac{6}{3}(x-3)^3 + \frac{4}{12}(x-9.5)^4 + C_1x + C_2$$

deflection

$$EI y = -2x^3 + 2(x-3)^3 + \frac{1}{3}(x-9.5)^4 + C_1x + C_2$$

Applying Bound. cond^s

at $(x, y) = (0, 0)$

$$EI y = -2x^3 + C_1x + C_2$$

$$EI \times 0 = -2 \times 0^3 + C_1 \times 0 + C_2$$

$$0 = C_2$$

at $(x, y) = (14, 0)$

$$EI y = -2x^3 + 2(x-3)^3 + \frac{1}{3}(x-9.5)^4 + C_1x + C_2$$

$$0 = -2 \times 14^3 + 2(14-3)^3 + \frac{1}{3}(14-9.5)^4 + 14C_1 + 0$$

$$C_1 = \underline{\hspace{2cm}}$$

Slope at B $(x, y) = (3, 0)$

$$EI \frac{dy}{dx} = -6x^2 + C_1 \quad \text{at } x=3, y=0$$

$$2 \times 10^5 \times 10^8 \frac{dy}{dx} = -6 \times 3^2 + 198.17$$

$$\frac{dy}{dx} = \frac{-54 + 198.17}{2 \times 10^5 \times 10^8}$$

deflection at B

$$EI y = -2x^3 + C_1 x + C_2$$

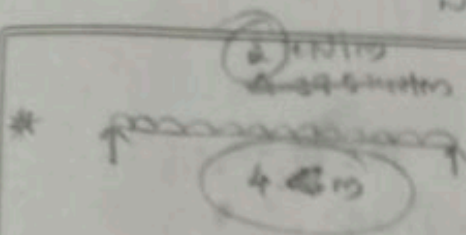
$$= -2 \times 3^3 + 198.17 \times 3 + C_2$$

$$y = \frac{-2 \times 3^3 + 198.17 \times 3 + C_2}{2 \times 10^5 \times 10^8}$$

$$y = 0$$

$$-2 \times 3^3 + 198.17 \times 3 + C_2 = 0$$

W/m

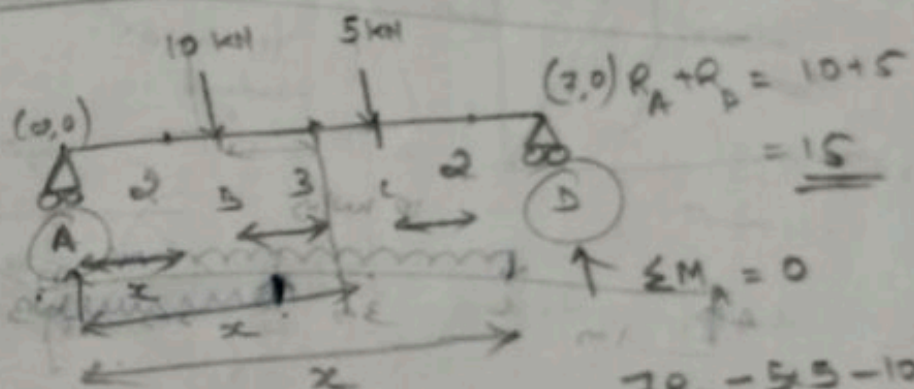


$$EI = 80 \times 10^9 \text{ Nmm}^2$$

$$y = 0.95$$

$$\sigma = 7 \text{ N/mm}^2$$

$$y = \frac{5 \omega l^4}{384 EI} = \frac{5 \times 2 \times (4000)^4}{384 \times 80 \times 10^9} = 73$$



$$(3.0) R_A + R_D = 10 + 5 = 15$$

$$\sum M_A = 0$$

$$7R_D - 5 \times 3 - 10 \times 2 = 0$$

$$7R_D - 25 - 20 = 0$$

$$7R_D = 45$$

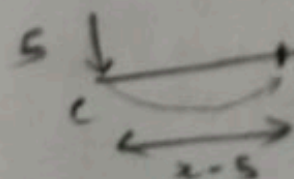
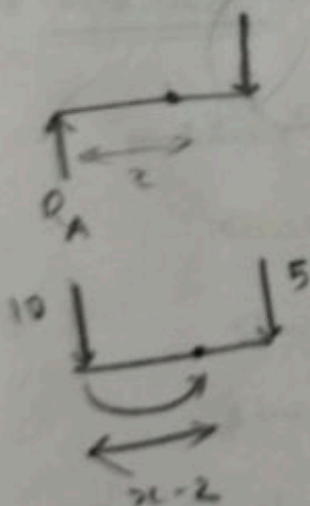
$$R_D = 6.4$$

$$R_A = 8.6$$

$$M_1 = -8.6 \times x$$

$$M_2 = 10 \times (x - 2)$$

$$M_3 = 5 \times (x - 5)$$

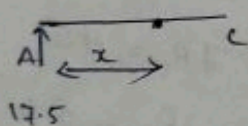
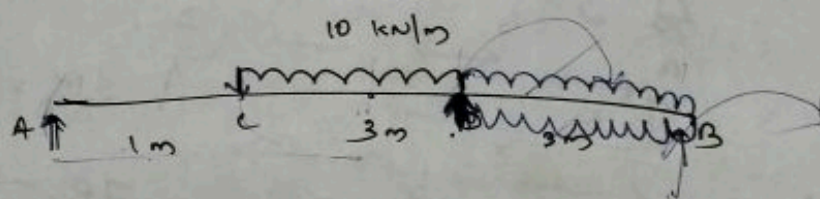


$$EI \frac{d^2 y}{dx^2} = M$$

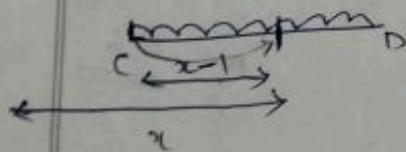
$$M = M_1 + M_2 + \dots$$

$$EI \frac{d^2 y}{dx^2} = -9.6x + 10(x-2) + 5(x-5) \quad \text{--- (1)}$$

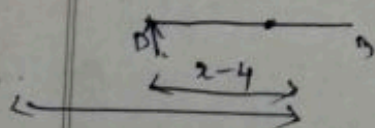
$$EI \frac{dy}{dx} = -\frac{9.6x^2}{2} + 10 \frac{(x-2)^2}{2} + 5 \frac{(x-5)^2}{2} + C_1$$



$$\Rightarrow M_1 = -17.5x$$



$$\Rightarrow M_2 = 10(x-1) \frac{(x-1)}{2}$$



$$\Rightarrow M_3 = -10(x-4) \frac{(x-4)}{2}$$

$$EI \frac{d^2 y}{dx^2} = -17.5x + 10 \frac{(x-1)^2}{2} - 10 \frac{(x-4)^2}{2}$$